

GPT-4o Refined Prompt Template for LLMatDesign

I have a material with a known **<property>**. **<definition of property>**.

Material information:

- Chemical formula: **<chemical composition>**
- **<property>**: **<property value>**

Objective:

Propose a modification to this material to achieve **<objective>**. You can choose one of the following modification types:

1. exchange: exchange two elements in the material
2. substitute: substitute one element in the material with another
3. remove: remove an element from the material
4. add: add an element to the material

Your response should be a Python dictionary in the following format:

```
'''
{Hypothesis: $HYPOTHESIS, Modification: [$TYPE, $ELEMENT_1, $ELEMENT_2]}.
'''
```

Requirements:

1. \$HYPOTHESIS: Provide a detailed analysis and rationale for your proposed modification.
2. \$TYPE: Specify the type of modification ("exchange", "substitute", "remove", "add").
3. \$Identify the element(s) involved in the modification. For "exchange" and "substitute", include two elements (\$ELEMENT_1 and \$ELEMENT_2). For "remove" and "add", include one element (\$ELEMENT_1).

<modification history>

Take a deep breath and work on this problem step-by-step. Your thoughtful and detailed analysis is highly appreciated.

Figure A.2: GPT-4o refined prompt template for LLMatDesign. Text placeholders in red angular brackets are specific to the task given to LLMatDesign. Text placeholders in blue angular brackets are optional and can be omitted if not needed.

Persona Prompt Template for LLMatDesign

You are a materials design expert working on the development of new materials with specific properties. You will be given a composition (chemical formula) and its corresponding **<property>**. You will be asked to propose a modification to the material to achieve a target **<property>**.

Material information:

- Chemical formula: **<chemical composition>**
- **<property>**: **<property value>**

Objective:

Propose a modification to this material to achieve **<objective>**. You can choose one of the following modification types:

1. exchange: exchange two elements in the material
2. substitute: substitute one element in the material with another
3. remove: remove an element from the material
4. add: add an element to the material

Your response should be a Python dictionary in the following format:

```
'''
{Hypothesis: $HYPOTHESIS, Modification: [$TYPE, $ELEMENT_1, $ELEMENT_2]}.
'''
```

Requirements:

1. \$HYPOTHESIS: Provide a detailed analysis and rationale for your proposed modification.
2. \$TYPE: Specify the type of modification ("exchange", "substitute", "remove", "add").
3. \$Identify the element(s) involved in the modification. For "exchange" and "substitute", include two elements (\$ELEMENT_1 and \$ELEMENT_2). For "remove" and "add", include one element (\$ELEMENT_1).

<modification history>

Take a deep breath and work on this problem step-by-step. Your thoughtful and detailed analysis is highly appreciated.

Figure A.3: Prompt template with materials design expert persona for LLMatDesign. Text placeholders in red angular brackets are specific to the task given to LLMatDesign. Text placeholders in blue angular brackets are optional and can be omitted if not needed.

B Convergence Plots

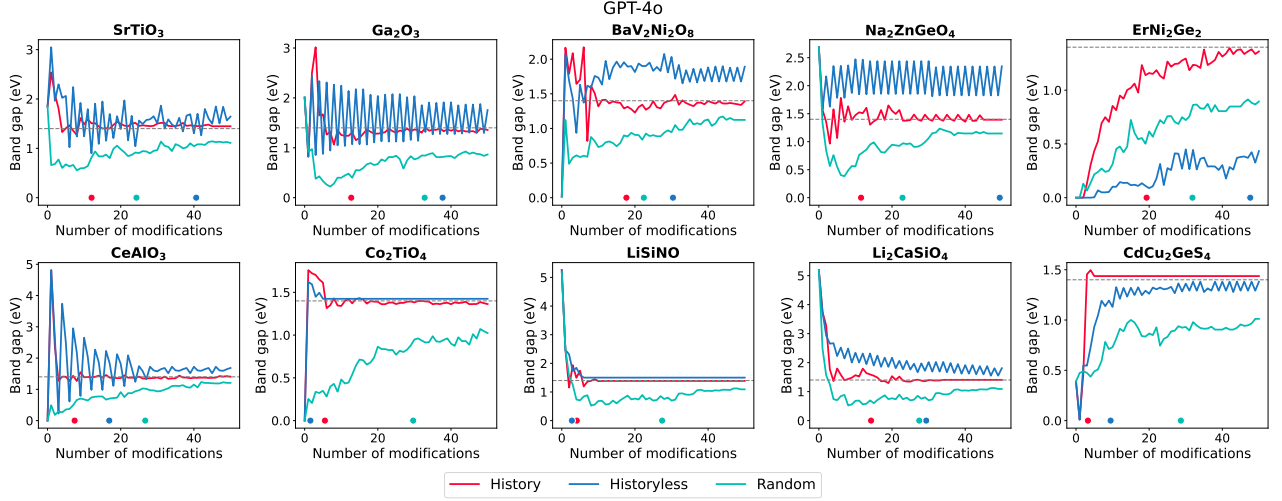


Figure B.1: Average band gaps over 50 modifications for all 10 starting materials using GPT-4o. The grey horizontal line indicates the target band gap of 1.4 eV. The colored dots on the x-axis indicate the average number of modifications taken for each method to reach the target.

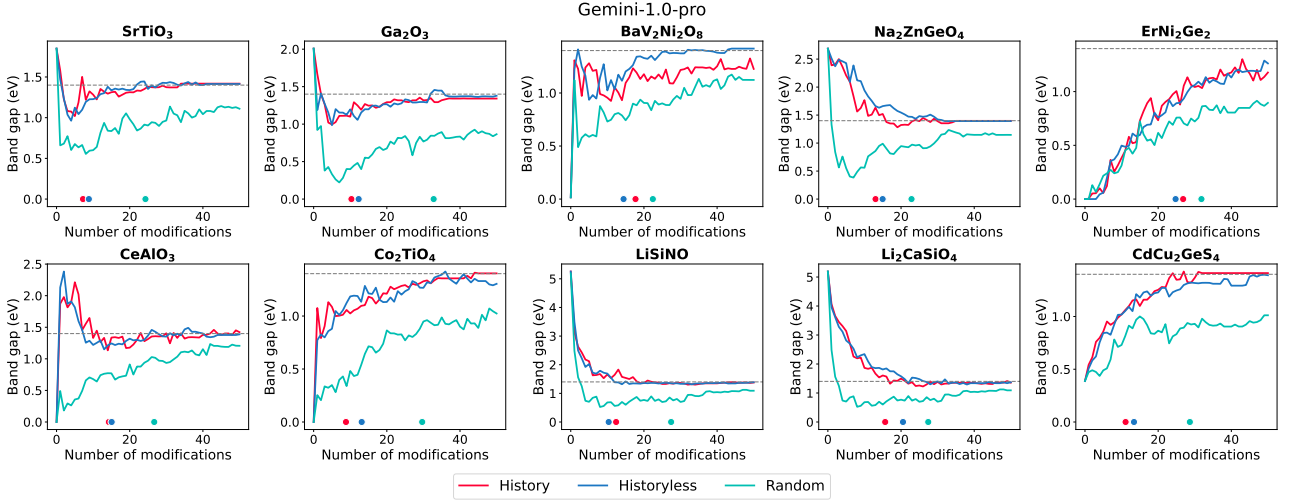


Figure B.2: Average band gaps over 50 modifications for all 10 starting materials using Gemini-1.0-pro. The grey horizontal line indicates the target band gap of 1.4 eV. The colored dots on the x-axis indicate the average number of modifications taken for each method to reach the target.

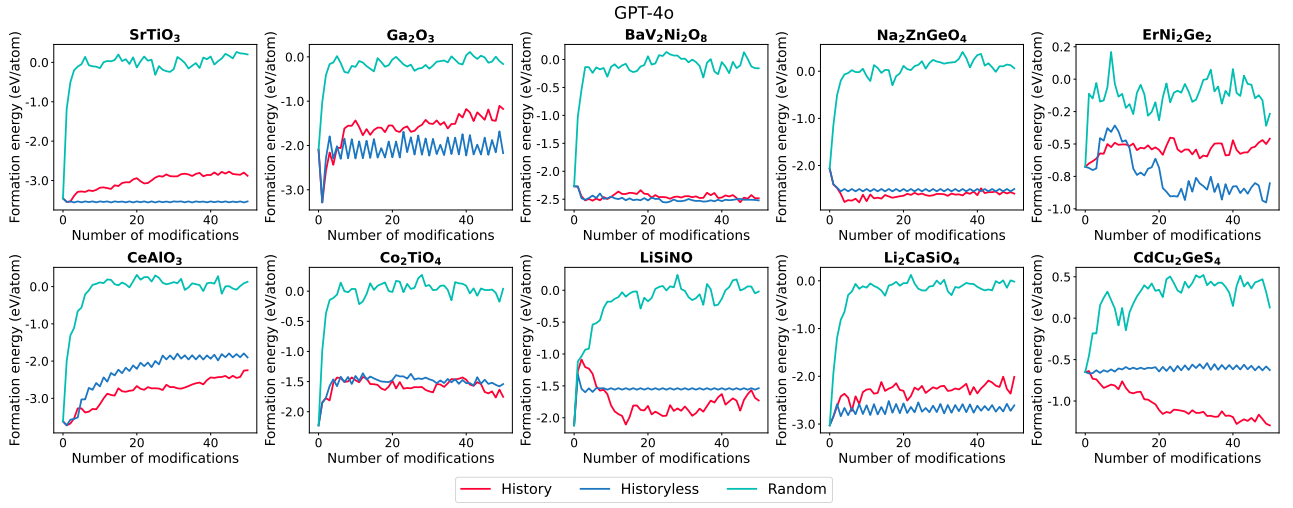


Figure B.3: Average formation energies over 50 modifications for all 10 starting materials using GPT-4o. The goal is to achieve the lowest possible formation energy per atom.

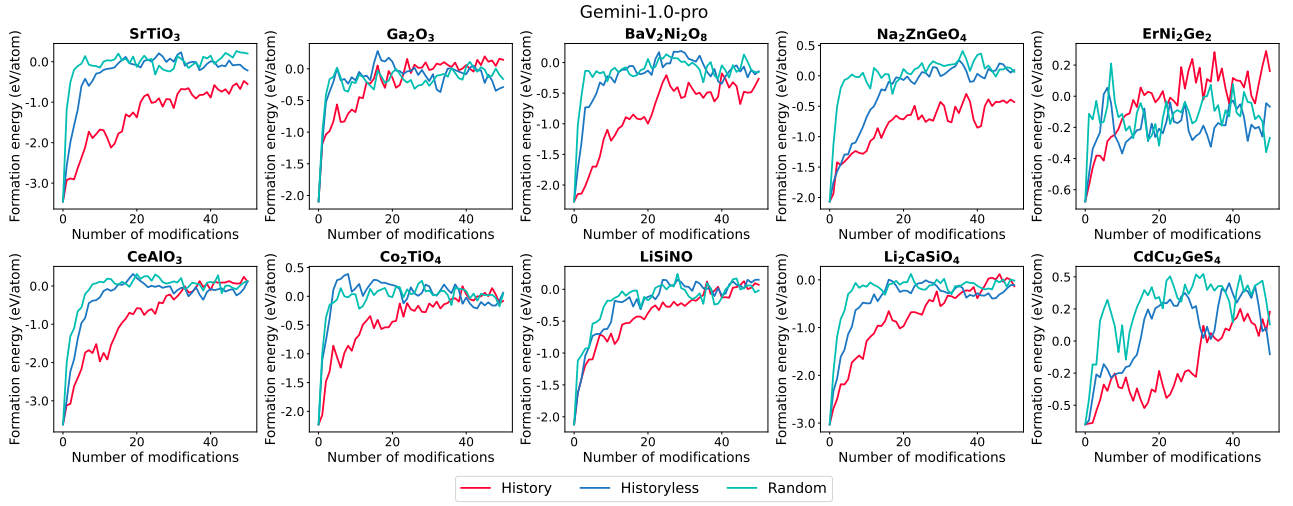
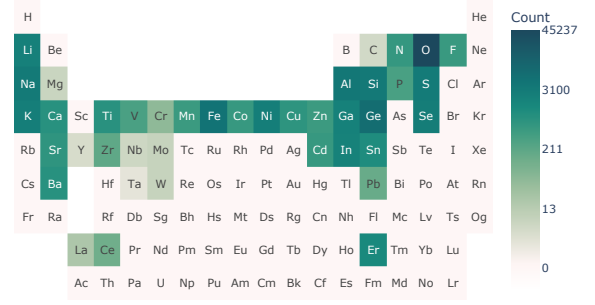
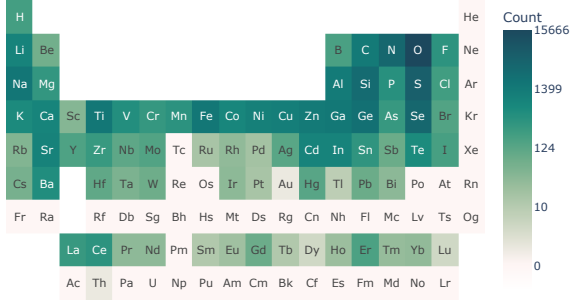
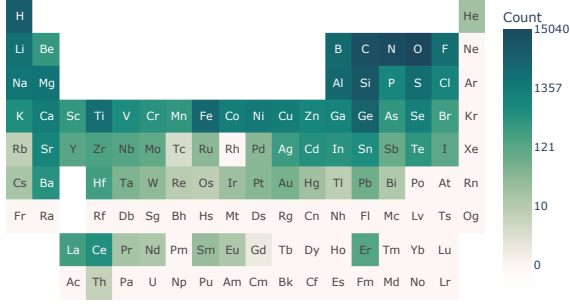


Figure B.4: Average formation energies over 50 modifications for all 10 starting materials using Gemini-1.0-pro. The goal is to achieve the lowest possible formation energy per atom.

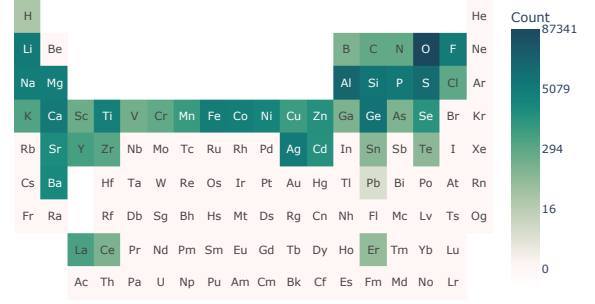
C Heatmaps



BG: Gemini-1.0-pro without history



BG: GPT-4o without history



FE: Gemini-1.0-pro without history

FE: GPT-4o without history

Figure C.1: Heatmaps of element frequencies in band gap (BG) and formation energy (FE) tasks for Gemini-1.0-pro and GPT-4o without history. The periodic table is color-coded to indicate the frequency of each element's occurrence in all modified materials (both intermediate and final) across all runs and starting materials. Darker colors represent higher frequencies, while lighter colors denote lower frequencies or absence. The visualization employs log-scaling to effectively highlight the distribution and prevalence of elements.

D Visualization of Selected Structures

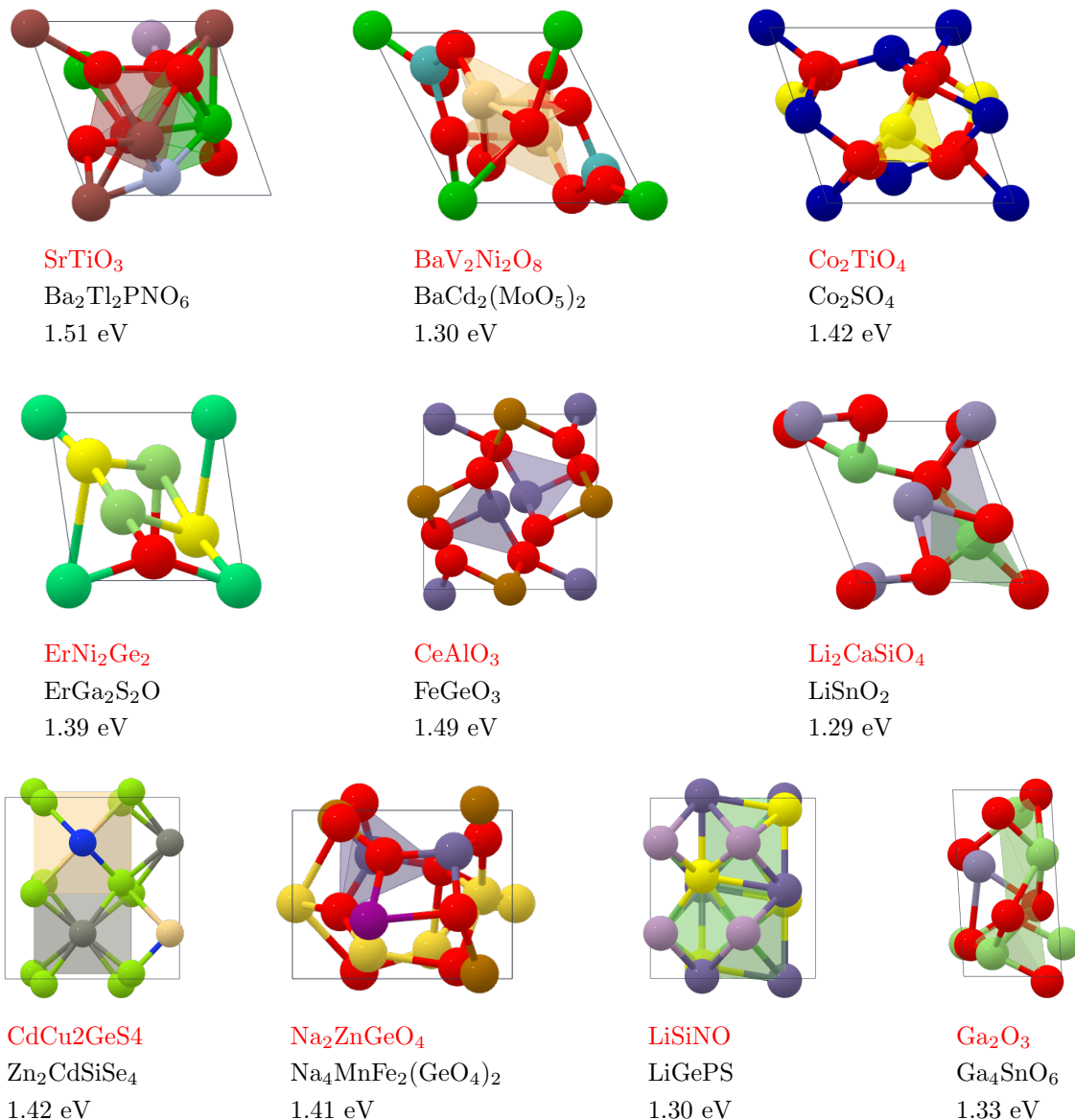


Figure D.1: Visualization of the final structures obtained by LLMatDesign for the band gap task. These structures are obtained from the first run of all 10 starting materials. The chemical formulae in red represent the starting materials, followed by the formulae of the final structures and their corresponding band gaps. GPT-4o with history is utilized as the LLM engine.